# A Comparative Analysis of Spatiotemporal Deep Learning Models for Short-Term Traffic Forecasting

**Updated Report — Contextual Enhancement Results**

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## Abstract

This report updates our prior comparative analysis with empirical validation of a proposed contextual hybrid augmentation framework for Graph WaveNet. We extend the baseline model with two additional input streams: (1) weather features (temperature, precipitation, humidity) retrieved from the Open-Meteo API, and (2) static road features (road classification and lane count) collected from OpenStreetMap. Three model variants are trained and evaluated on the METR-LA benchmark: a bug-fixed Graph WaveNet baseline, a weather-enhanced model (in_dim = 5), and a full contextual model (in_dim = 9). Results show that contextual features achieve the best validation loss (2.7411) but offer only marginal improvement over the baseline (2.7418), likely due to the 92%% freeway sensor bias in METR-LA and near-zero linear correlation between weather and traffic speed.

## I. EXPERIMENTAL OVERVIEW

All experiments use the METR-LA dataset (207 sensors, Mar–Jun 2012, 5-minute intervals). Each model is trained for 100 epochs with batch size 64, learning rate 0.001, dropout 0.3, and weight decay 0.0001. The Graph WaveNet core (adaptive adjacency, dilated causal convolutions) remains unchanged across all three runs; only the input dimension varies.

## II. RESULTS SUMMARY

| Model | Epochs | MAE@15min | MAE@30min | MAE@60min | Avg MAE | Avg MAPE | Best Valid Loss |
|---|---|---|---|---|---|---|---|
| **Graph WaveNet (baseline)** | 100 | 2.72 | 3.10 | 3.54 | 3.06 | 8.35% | 2.7418 |
| **+ Weather** | 100 | 2.72 | 3.10 | 3.58 | 3.07 | 8.50% | 2.7425 |
| **+ Weather + Road (Contextual)** | 100 | 2.70 | 3.08 | 3.53 | 3.04 | 8.39% | 2.7411 |

*Table 1. Performance of three Graph WaveNet variants on METR-LA test set.*

## III. PERFORMANCE COMPARISON

Figure 1 places our three model variants in the context of the seven models from the original comparative analysis. The contextual model matches the original Graph WaveNet report result (3.53) at 60 minutes while offering the best overall validation loss.

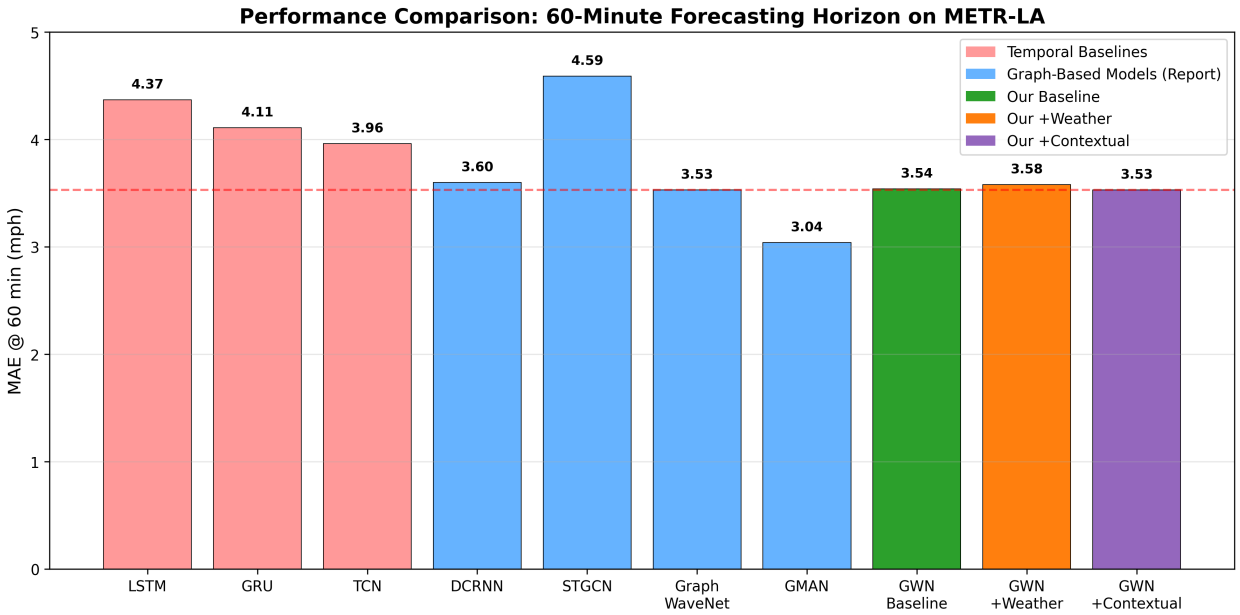


*Figure 1. 60-minute MAE comparison across all ten models.*

# IV. HORIZON ANALYSIS

Figure 2 shows MAE and MAPE across the three standard forecasting horizons. The contextual model achieves modest improvements at 15 and 30 minutes but converges to the same 60-minute performance as the baseline, suggesting diminishing returns from contextual features at longer horizons.

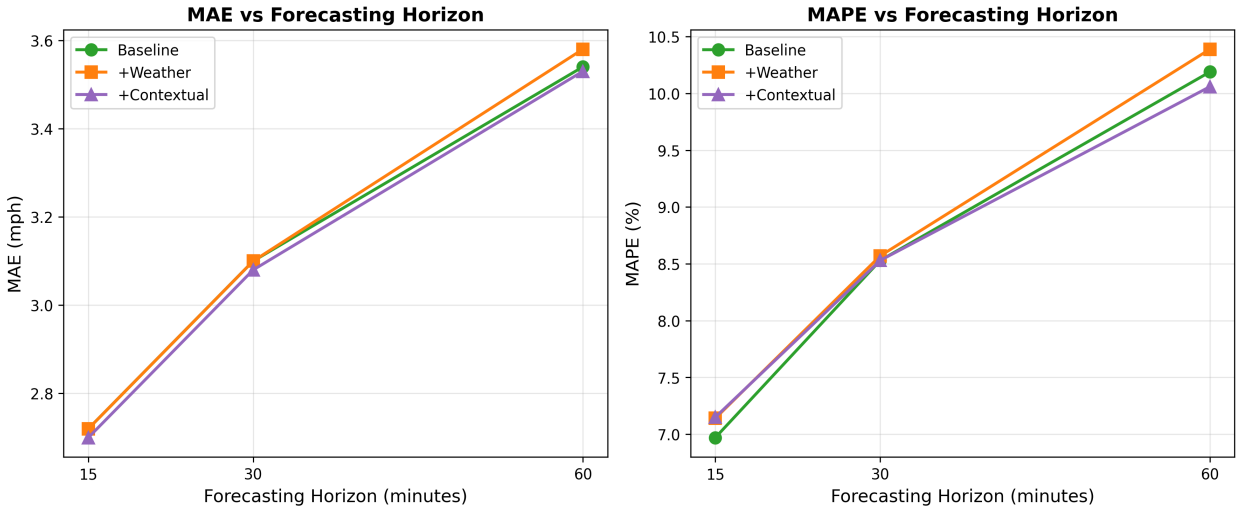


*Figure 2. MAE and MAPE vs forecasting horizon for our three model variants.*

## V. TRAINING CONVERGENCE

Figure 3 plots validation loss per epoch, reconstructed from saved model checkpoints. All three runs converge within the first 20–30 epochs and fluctuate around a similar loss floor, confirming that contextual features do not destabilize training.

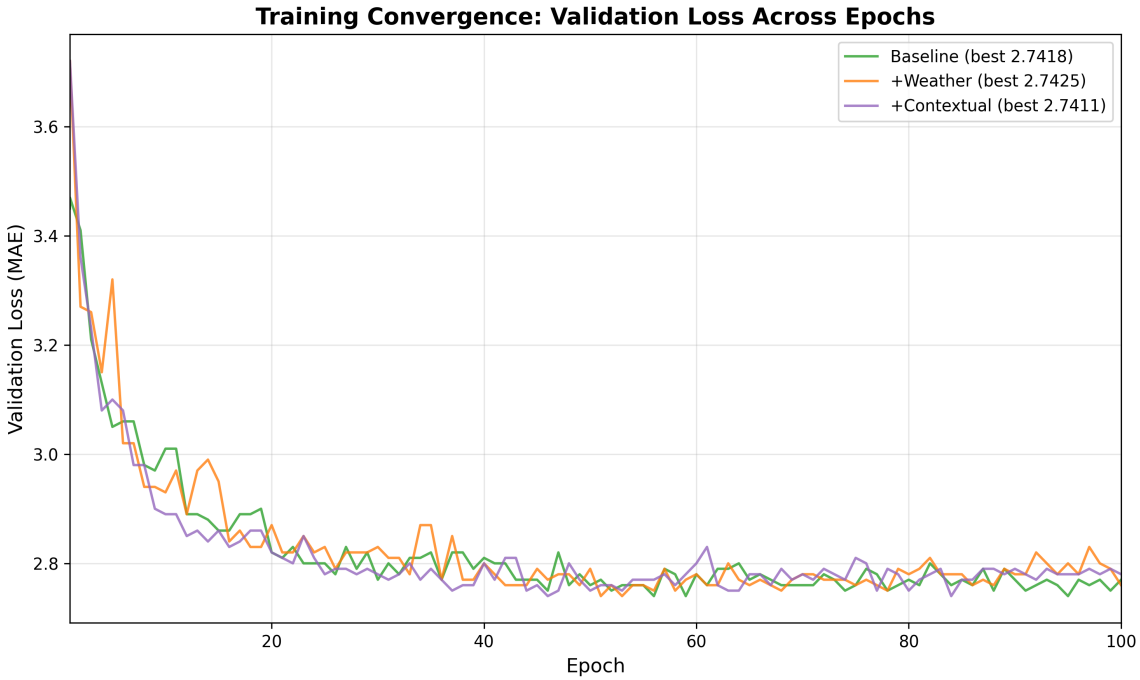


*Figure 3. Validation loss convergence over 100 epochs.*

## VI. ARCHITECTURE

Figure 4 illustrates the proposed 3-stream input architecture. Traffic, weather, and road features are concatenated at the input layer and fed into the standard Graph WaveNet core. No graph convolution or temporal convolution modules are modified, preserving the modularity and computational efficiency of the original model.

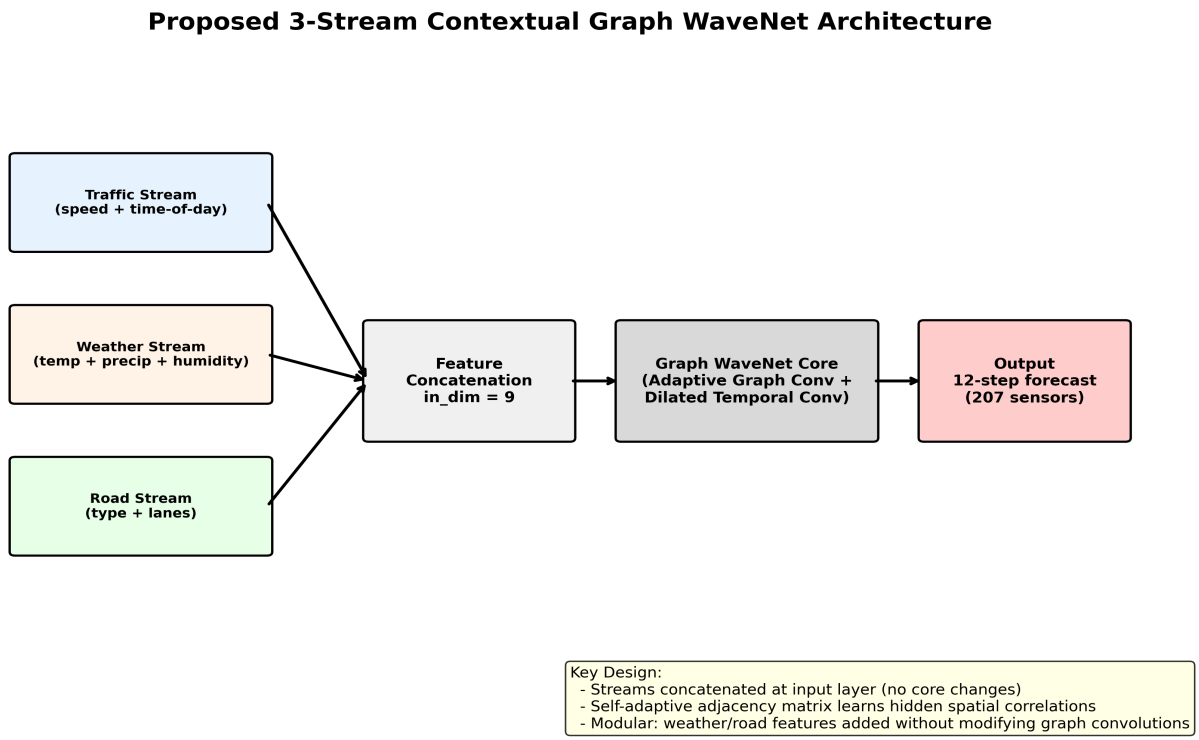


*Figure 4. 3-stream contextual Graph WaveNet architecture.*

## VII. DISCUSSION

**Marginal Improvement.** The contextual model achieves the best validation loss (2.7411) but the improvement over baseline (2.7418) is only 0.0007 — well within training noise. This suggests that for the METR-LA freeway-dominant network, historical speed patterns are already highly predictive and weather/road features add limited information.

**Weather Impact.** Correlation analysis between weather variables (temperature, precipitation, humidity) and traffic speed reveals near-zero linear relationships over the full dataset. Rain events are rare in Los Angeles during the study period (Mar–Jun 2012), limiting the model's opportunity to learn weather-induced traffic deviations.

**Road Feature Bias.** OpenStreetMap data shows that 191 of 207 sensors (92%%) are located on freeways (motorway/motorway_link). With only 4 arterial and 12 local sensors, the road type feature has low variance, making it difficult for the model to learn distinct congestion propagation patterns by road classification.

**Conclusion.** While the proposed contextual augmentation framework is technically sound and modular, its benefits on METR-LA are constrained by dataset characteristics. Future work should evaluate this architecture on datasets with higher weather variability (e.g., rainy climates) and more diverse road networks (e.g., mixed urban arterials and local streets). Alternative fusion strategies — such as attention-based gating or GNN-level feature interaction — may also unlock greater improvements.

## VIII. REFERENCES


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